02-Inter-poly-exc

Horner

1. Use Horner scheme to evaluate by hand the polynomial e263cd

$$p(z) = z^5 + (1+2i)z^4 + (2-i)z^3 + (3-2i)z^2 + iz + \beta$$

at the point $z_0 = 1+i$, and then choose $\beta \in \mathbb{C}$ so that $p(z_0) = 0$.

2. Use Horner scheme to evaluate by hand the polynomial e263ce

$$p(z) = z^5 + (3-i)z^4 + 2iz^3 + (2+i)z^2 + (1-i)z + \beta$$

at the point $z_0 = 1+i$. Can $\beta \in \mathbb{C}$ be chosen so that z_0 becomes a fix point for p, that is, $p(z_0) = z_0$?

3. Let $p(z) = z^3 - (1+4i)z^2 - 5(1-2i)z + (17-6i)$. 8f963b

   (a) Use the Horner scheme to compute $p(2-i)$ and $p(1+2i)$.

   (b) Find the roots z_1, z_2, z_3, and factor the polynomial $p(z) = (z-z_1)(z-z_2)(z-z_3)$.

4. Use Horner to evaluate $p(z) = z^3 - 6z^2 + (14+3i)z - (15+5i)$ at $z = 2$, $z = 2-i$ and $z = 1+2i$. Do you have enough info to find all the roots of, and hence, factor $p(z) = (z-z_1)(z-z_2)(z-z_3)$? If yes, do so! d2120b

5. Use Horner to evaluate $p(z) = z^3 - 2z^2 - 5iz - 7 + i$ at $z = 2$, $z = i$ and $z = 3+i$. Do you have enough info to find all the roots of, and hence, factor $p(z) = (z-z_1)(z-z_2)(z-z_3)$? If yes, do so! d2120a

6. Consider the polynomial $p(z) = z^3 - (1+5i)z^2 - (3+2i)z + (17-i)$. f8f588

   (a) Use the Horner scheme to compute by hand the values of $p(1-i)$ and $p(-2+i)$.

   (b) Find all the roots z_1, z_2 and z_3 of p, so that $p(z) = (z-z_1)(z-z_2)(z-z_3)$.

7. The polynomial $p(x)$ below has all its roots $x_{1,2,3,4} \in \mathbb{Z}$. Use the Horner scheme to find quickly its roots and factor it, $p(x) = (x-x_1)(x-x_2)(x-x_3)(x-x_4)$, if 0e36f2

$$p(x) = x^4 - 5x^3 - 7x^2 + 29x + 30.$$

Interpolation/Divided differences

1. Use divided differences to construct the Newton interpolation polynomial $p(x)$ the graph of which passes through the points with coordinates $(-2,-32)$, (1,7), (3,23) and (4,46). Rewrite it afterwards in standard basis and check, that, e.g., $p(1) = 7$. f9f3f0

2. Let $y = p(x)$ be the interpolation polynomial through the nodes: 33ca8d

x_i	-1	1	3	4
y_i	-10	-2	14	40

   (a) Write down $p(x)$ in both Lagrange and in Newton form basis.

   (b) Expand the Newton form of $p(x)$ to obtain it in standard basis, $p(x) = a_3x^3 + a_2x^2 + a_1x + a_0$.

   (c) Check that the senior coefficient of the IP in Lagrange form is equal to a_3.

3. Let $y = p(x)$ be the interpolation polynomial through the nodes: 88193d

x_i	-1	1	2	3
y_i	-1	7	8	15

   (a) Write down $p(x)$ in both Lagrange and in Newton form basis.

   (b) Expand the Newton form of $p(x)$ to obtain it in standard basis, $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$.

   (c) Check that the senior coefficient of the IP in Lagrange form is equal to a_3.

4. Compose the interpolation polynomial (IP)through the points $(1,-15)$, $(2,-1)$, $(4,3)$, $(5,17)$: d85487

   (a) in Lagrange and in Newton form basis.

(b) Use the Newton form of the IP to write it in standard basis, $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$.

(c) Check that the senior coefficient of the IP in Lagrange form is equal to a_3.

5. Compose the interpolation polynomial (IP)through the points $(-1, 6)$, $(0, 5)$, $(1, 4)$, $(2, 15)$: 3f5034

   (a) in Lagrange and in Newton form basis.

   (b) Use the Newton form of the IP to write it in standard basis, $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$.

   (c) Check that the senior coefficient of the IP in Lagrange form is equal to a_3.

6. Linear interpolation between (a, f_a) and (b, f_b) are tabulated using weighted divided differences (wDD): bb4c48

$$\frac{a \quad x \quad b}{f_a \quad ? \quad f_b} \rightsquigarrow \lfloor \boldsymbol{\delta}{:}\mathbf{f} \rfloor := \left\{ \begin{array}{c|cc} \delta_a & f_a & \lfloor f_a{:}f_b \rfloor \\ \delta_b & f_b & \end{array} \right\}, \quad f_x = \lfloor f_a{:}f_b \rfloor = \frac{1}{\delta_a - \delta_b} \begin{vmatrix} \delta_a & f_a \\ \delta_b & f_b \end{vmatrix}, \quad \begin{cases} \delta_a = x - a, \\ \delta_b = x - b. \end{cases}$$

Show the following properties (λ below is a number):

$$(a)\ \lfloor \lambda\boldsymbol{\delta}{:}\mathbf{f} \rfloor = \lfloor \boldsymbol{\delta}{:}\mathbf{f} \rfloor,\ \lambda \neq 0, \quad \text{while} \quad \lfloor \boldsymbol{\delta}{:}\lambda\mathbf{f} \rfloor = \lambda \lfloor \boldsymbol{\delta}{:}\mathbf{f} \rfloor \qquad (b)\ \lfloor f_a{:}f_b \rfloor \overset{T}{=} \frac{1}{b-a} \begin{vmatrix} \delta_a & \delta_b \\ f_a & f_b \end{vmatrix}.$$

Take $\lambda = 100$ and exemplify (a) and (b) by interpolating in the table: $\dfrac{0.06 \quad 0.075 \quad 0.08}{2.4 \quad ? \quad 7.6}$.

(c) Finish by verifying that wDD actually performs linear interpolation:

$$\lfloor f_a{:}f_b \rfloor = \begin{vmatrix} \delta_a & f_a \\ -1 & \dfrac{f_b - f_a}{b-a} \end{vmatrix} = f_a + \frac{f_b - f_a}{b-a}(x - a).$$

7. Runge's phenomenon ... 7832a7

Parallelized

1. Below you find the table for the cumulative distribution functionof the standard Normal distribution: 6ea926

x	0.20	0.21	0.22	0.23	$\cdots$
$\Phi(x)$	0.57926	0.58317	0.58706	0.59095	$\cdots$

   Interpolate in the table using Neville's scheme to approximate $\Phi(0.216)$

   (a) linearly; (b) extend to a cubic using 4 nodes, two on each side

   Find the absolute error in both cases if the actual value of $\Phi(0.216) = 0.5855061262175560\ldots$.

2. Use divided differences to construct the Newton interpolation polynomial $p(x)$ with graph through the points $(-2, -2)$, $(-1, -6)$, $(0, -4)$ and $(1, -2)$. Compute then the value of $p(2)$ in two ways: 6b2927

   (a) by substitution in the Newton IP; (b) From the node values, using wDD/Neville's scheme.

   Finally, write down $p(x) = ax^n + \cdots + a_1x + a_0$ in standard basis.

Octave/Python

1. Devise a routine for computing weighted divided differences $\lfloor \boldsymbol{\delta}:\mathbf{f} \rfloor$, taking $\{(\delta_i, f_i)\}_{i=0}^{n}$ and $0 < j < n$ as input parameters, and outputs the j:th column of the table $\{\lfloor f_i:f_{i+j} \rfloor\}_{i=0}^{n-j}$ as a list, e.g. (Python): 1455a0

```
import numpy as np
def wDD(di,yi,col=100):  # Scalar version
    n=len(di); lastcol = min(n,col+1)
    Y = np.array(yi)    # Init return array
    for j in range(1,lastcol):
        for i in range(n-j):
            Y[i] = (di[i]*Y[i+1] - di[i+j]*Y[i]) / (di[i] - di[i+j])
    return Y[:n-lastcol+1]
```

Test by, say, Neville-evaluating the IP $p(x) = x^3$ at $x = 5$ (last line reads $5^3 = 125$ as it should?):

```
n=4; xi=np.arange(n); yi=xi**3;
for j in range(n):
    print(f"# >> Column {j}: {wDD(5-xi,yi,j)}")
```